

# Why Do People Resist Drone Food Delivery Services? An Innovation Resistance Theory Perspective

Ashraf Khalil , Amit Shankar , Rahul Bodhi , Abhishek Behl , and Alberto Ferraris 

**Abstract**—Recently, the food service delivery industry started using drones for delivering food. The drone food delivery services are not only beneficial for marketers but also provide several benefits to the consumers. Despite several advantages for different stakeholders, the adoption of drone-based delivery is not encouraging. Thus, it is crucial to know why consumers show resistance behavior toward drone-based food delivery services. Toward that end, this study examines the effects of different functional and psychological barriers on consumers' resistance to drone-based food delivery services through the lens of innovation resistance theory. A moderated-mediation framework has been proposed using anticipated negative emotions as a mediator and consumers' innovativeness as a moderator. The hypotheses are examined by analyzing the 355 samples collected from online food delivery service users. The results suggest that usage, risk, image, and experience barriers are significantly associated with resistance. The study's findings enrich emerging drone-based delivery services and online food delivery literature. The study's findings furthermore offer practical knowledge to help online food delivery service providers implement strategies to reduce consumers' resistance to the drone food delivery system.

**Index Terms**—Anticipated negative emotions, consumer innovativeness, drone-based food delivery, innovation resistance theory (IRT), online food delivery.

## I. INTRODUCTION

**D**UE to the increased utilization of drones in various industries, such as e-commerce delivery, agriculture, defence, public utility services, entertainment, and broadcasting, delivery of goods by drones is experiencing growth and popularity across the globe [1], [2]. The current market share of the drone delivery

industry is approximately USD 528 million and is expected to reach USD 39 013 million by 2030, with an annual growth rate of 53.8% [3]. Further, COVID-19 has contributed to the growth of the drone delivery industry as consumers now prefer contactless delivery [4]. The food service delivery industry is not an exception, and marketers have started using drones for food delivery [1]. Online food delivery has witnessed significant growth with an annual growth rate of 9.6% and is predicted to reach USD 137596 million by 2023 [5]. Drone food delivery services are not only beneficial for marketers but also provide several benefits to the consumers [6].

The food delivery industry is the first mover in technology adoption for delivering food to consumers in a cost-effective manner [1]. Hence, in the recent past, the industry has witnessed several technological advancements. For instance, restaurants started connecting with consumers by developing their own website. Further, smartphone penetration motivated restaurants to connect with consumers using mobile application platforms [1]. However, developing and managing websites is a costly affair. As a result, third party aggregators emerged to manage online orders and deliver to consumers in a cost-effective manner with limited liability. However, delivery on demand by humans is not environmentally friendly, especially given its rapid increase in popularity. Consequently, eco-friendly restaurants initiated delivery by drones [8].

In the online food delivery context, drone delivery also reduces delivery time due to the avoidance of traffic congestion and is also cost-effective as it does not require human intervention to deliver the food [7]. Hence, several food delivery companies started using drones to deliver food quickly and effectively [8]. Despite several benefits, the adoption of drone-based food delivery is not as expected as consumers have established high expectations regarding food delivery and are skeptical about the performance of drone-based food delivery services [9]. The absence of human touch is another reason for consumers' resistance [10]. Hence, marketers are keen to know why consumers are not demonstrating positive intent toward drone-based food delivery systems.

Researchers have begun to explore this issue, with existing studies focused on investigating factors affecting consumers' intention to adopt drone-based food delivery [1], [6], [7], [8]. Studies have highlighted the role of the public image of drone food delivery services [9], environmental locus of control [8], psychological benefits [6], attitudes [10], and subjective norms

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[1] in drone-based food delivery system adoption. Broadly, existing literature has highlighted various factors which have positive/negative impact on consumers' adoption of drone-based food delivery systems [1], [6]. Yet, little effort has been made to investigate intention to resist drone delivery. This represents a literature gap. Consumer resistance refers to consumers' unwillingness or unfavorable response toward adoption and use of an innovative system [11]. Consumers resistance is an essential element of consumer behavior and plays a crucial role in framing consumers' response toward specific phenomena [12]. Hence, consumer resistance has been investigated with regard to mobile payment [11], electric vehicles [12], online food delivery [13], mobile HRM systems [14], emerging technologies [15], [16], digital innovation [17], retail investment [18], organic food buying [19], open innovation [20], food delivery [21], AI-enabled voice assistance [22], artificial intelligence [23], and healthcare [24], but it has not been investigated in the drone-based food delivery context. Further, innovation resistance theory (IRT) found to be one of the predominant theories in investigating resistance behavior. IRT emphasized how different functional and psychological risks play a role in framing consumers' resistance behavior. Hence, to fulfil this gap, this study investigates the effects of various barriers on consumers' resistance to drone-based food delivery using IRT as underpinning theory.

Moreover, literature has highlighted that consumers' resistance behavior is different from consumers' adoption behavior [14]; hence, factors which drive resistance are expected to be different from factors which motivate consumers to adopt it [13]. Therefore, for deeper understanding of consumer behavior, it is warranted to investigate factors affecting consumers' resistance. This study uniquely contributes to the existing drone-based food delivery literature by examining the role of barriers in framing consumers' intentional resistance.

Further, emotions play a crucial role in framing consumers' perception of new technology [25]. Consumers represent positive or negative emotions toward new technology as per their perception, which leads to behavioral intention. As such, it is critical to understand how different barriers help in framing consumers' negative emotions toward drone-based food delivery and, ultimately, consumers' behavioral intention. Thus, this study attempts to investigate the mediating effect of anticipated negative emotions between different barriers and resistance intention toward drone-based food delivery.

In addition to emotions, consumers' individual level of innovativeness plays a crucial role in framing their response to new technology [14]. Reflecting its importance, the existing literature includes studies investigating the effect of personal innovativeness on consumer behavioral intention in the contexts of mobile banking [26], food delivery [6], and mobile payment [13]. The current study goes a step further, exploring the effect of innovativeness on consumers' resistance behavior. This study attempts to fill the gaps mentioned above by empirically investigating consumers' resistance to drone-based food delivery through the lens of innovation resistance theory. Specifically, this study attempts to answer the following research questions (RQs):

RQ1: Why do consumers exhibit resistance behavior toward drone-based food delivery?

RQ2: How do anticipated emotions play a role in framing consumers' resistance behavior?

RQ3: Does consumers' resistance to drone-based food delivery vary based on consumers' innovativeness?

This study proposes a moderated-mediation framework to answer the above RQs. To answer RQ1, this study investigates how various barriers including usage barriers, value barriers, risk barriers, tradition barriers, image barriers, and experience barriers frame consumers' resistance to drone-based food delivery. Furthermore, the study investigates the mediating effects of anticipated negative emotion and how consumers' innovativeness moderates the mediating effect of anticipated negative emotions to answer RQ2 and RQ3, respectively.

This study involves several theoretical and practical contributions. Theoretically, this study makes contributions to food delivery literature, especially in the drone delivery category, by investigating consumers' resistance. Furthermore, aside from the limited number of studies using IRT in connection to online food delivery, to the best of authors' knowledge, no study has used IRT in the drone-based food delivery context. Hence, this study extends the IRT literature by exploring it in a new context. Moreover, in the existing literature, while attempts have been made to investigate consumers' perception toward drone-based food delivery in the developed market, scant efforts have been made to investigate consumers' perception and behavior in emerging markets. Food delivery unicorn Swiggy started trials for drone-based food delivery and conducted over 300 drone-led deliveries of food in India [3]. The literature has yet to catch up to such developments as they spread worldwide.

This study seeks to fill that gap, not only pertaining to emerging markets but also with respect to geographic scope, investigating consumers' resistance to in a lucrative emerging market, namely, drone-based food delivery in India. In the context of online food delivery, India offers a substantially large untapped market, which marketers are keen to penetrate using advanced technology. However, to the best of the authors' knowledge, no study has investigated Indian consumers' behavioral intention toward drone-based food delivery services. This study enriches the existing literature by investigating potential barriers to drone-based food delivery services in the Indian market. The study further adds to the literature an examination of the mediating effect of anticipated negative emotions and how this mediating effect varies given different levels of innovativeness among consumers. As consumer behavior is a complex phenomenon, the effect of different barriers on consumers' resistance intention is not straight-forward. Hence, it is crucial to investigate the role of intervening variables. This study uniquely contributes to existing literature by examining the dynamic roles of anticipated negative emotions (mediator) and consumers' innovativeness (moderator) in framing consumers' unfavorable perception toward drone-based food delivery services. Finally, in addition to the multiple significant theoretical contributions, this study provides practical guidance to restaurants and food delivery service providers in developing innovative strategies to

TABLE I  
IRT LITERATURE

| Author/Year               | Context                       | Method            | Country             | Barriers                                                                               | Dependent Variables                    |
|---------------------------|-------------------------------|-------------------|---------------------|----------------------------------------------------------------------------------------|----------------------------------------|
| Sivathanu (2018)          | Digital payment               | Survey-based      | India               | Usage, Value, Risk, Traditional, Image                                                 | Actual Usage                           |
| Kaur et al. (2020)        | Mobile payment                | Survey-based      | India               | Usage, Value, Risk, Traditional Image                                                  | Use intention, Intentions to recommend |
| Talwar et al. (2020)      | Online travel agencies        | Mixed method      | India               | Usage, Privacy and security, Vulnerability, benefits                                   | Purchase intention                     |
| Kushwah et al. (2019)     | Consumption of Organic Food   | Survey-based      | India               | Usage, Value, Risk                                                                     | Choice behaviour                       |
| Kaur et al. (2021)        | Food delivery applications    | Mixed method      | India               | Usage, Value, Risk, Traditional, Image (customer service), Image (Customer experience) | Intention to use, Word of mouth        |
| Joachim et al. (2018)     | Mobile services               | Survey-based      | Germany             | Usage, Value, Image                                                                    | Adoption Intention                     |
| Oktavianus et al. (2017)  | Mobile gaming                 | Qualitative study | Taiwan              | Usage, Value, Risk, Traditional, Image                                                 | Adoption Intention                     |
| Yu and Chantatub (2016)   | Mobile banking                | Survey-based      | Thailand and Taiwan | Usage, Value, Risk, Traditional, Image                                                 | Resistance intention                   |
| Chen et al. (2022)        | Mobile ticketing applications | Survey-based      | Taiwan              | Usage, Value, Risk, Traditional, Image                                                 | Intention to use                       |
| Chakraborty et al. (2022) | Fitness application           | Survey-based      | India               | Usage, Value, Risk                                                                     | Purchase behaviour                     |
| Kumar et al. (2022)       | Home service applications     | Survey-based      | India and USA       | Value, Risk, Traditional, Image                                                        | Trust and Word-of-Mouth                |

reduce consumers' negative emotions and resistance behavior toward drone-based delivery.

## II. UNDERPINNING THEORY AND LITERATURE REVIEW

### A. Innovation Resistance Theory (IRT)

IRT [27] is one of the prominent frameworks for investigating consumers' resistance to technology (see Table I). According to Ram and Sheth [27], consumers' perceived barriers play a crucial role in establishing consumers' perception toward existing technology and resistance to the adoption of new technology. IRT suggests that functional and psychological barriers are two significant barriers that frame consumers' negative intentions toward new technology [28]. The functional barriers mainly relate to technology as active barriers [17], whereas psychological barriers, which are related to individuals, have a passive effect on behavioral intention [13]. We have selected IRT as the underpinning theory in this study as IRT is the best-suited framework for investigating the effect of barriers on consumers' resistance intention. Further, IRT has been used in the existing literature to investigate consumers' resistance in various contexts [13], [29], [30].

In this study, usage barriers, value barriers, and risk barriers are proposed as functional barriers, while tradition barriers, image barriers, and experience barriers are proposed as psychological barriers that affect consumers' intentions toward drone-based food delivery. Usage barriers relate to consumers' apprehension toward usage of innovation [13]. If consumers perceive that the benefits received from innovation are less compared to the costs associated with it, they perceive less value in adoption

and more value in resistance to the innovation [14], hence value barriers. Risk barriers are associated with consumers' feeling of uncertainty and risk involving an innovation. Traditional barriers arise due to the status quo of consumers' perception toward established culture [17]. Image barriers are associated with the image of the innovative brand or industry [27]. Finally, experience barriers are consumers' uncertainties toward how they will feel when they actually use the specific innovation [18]. IRT is considered to be the most suitable underpinning theory given its extensive use in e-commerce literature to investigate resistance intention [17], which helps in understanding consumers' perceived apprehension to drone-based food delivery services.

### B. Drone-Based Food Delivery

Food delivery services have witnessed rapid technological advancement, resulting in greater effectiveness and efficiency [19]. Accordingly, online and mobile application platforms have largely replaced offline channels for food delivery [31]. Drones are one of the innovative delivery tools used for delivering food, which is more effective in terms of cost, time, and convenience [32]. Drones are small, sensor-equipped aircraft with automatic flying capacity [33]. Drones have been majorly used for event management and more recently by e-commerce companies for delivering products quickly [34]. Most recently, food delivery service providers, such as Domino's Pizza and Yogiyo have started using drones to deliver food items [10], [35]. Using a drone is cost-effective for food delivery service providers and enables them to deliver the food quickly, anytime, and anywhere [8]. Consequently, companies can extend their delivery reach to remote areas where limited or no ground transportation infrastructure is available [36].

Moreover, drone delivery is considered an environment-friendly tool as drones require significantly less fuel to deliver the food than traditional delivery methods [1], [34]. Consumers also find drone-based food delivery more hygienic and safer [7]. Drone delivery also avoids and helps in reducing traffic jams and road accidents [37]. Despite these many benefits, some consumers are skeptical toward drone-based food delivery due to the absence of human touch [9]. Consumers also hold negative perceptions toward drone food delivery because of privacy and security risks associated with it [8]. Further, some consumers avoid using drone services as they perceive that it would replace human labor, which may lead to unemployment, especially in the gig economy [38]. Hence, it is interesting to investigate how various barriers enhance consumers' resistance intention toward drone food delivery.

## III. RESEARCH METHODOLOGY AND HYPOTHESES DEVELOPMENT

Using IRT as the underpinning theory, we have proposed usage barriers, value barriers, and risk barriers as functional barriers and tradition barriers, image barriers, and experience barriers as psychological barriers in investigating the effect of these barriers on resistance intention. (See Fig. 1.)

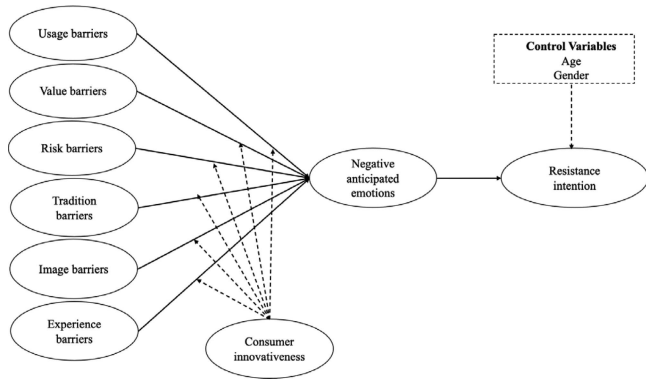


Fig. 1. Proposed framework.

### A. Usage Barriers

The usage barrier is one of the significant barriers in the new technology adoption process [27]. Individuals demonstrate resistance toward new technology as they are not familiar with the performance of the new technology [27]. Drone food delivery is a new system; hence, consumers cannot predict performance before use [9]. Further, individuals perceive that learning to use new technology requires effort and changing their routine [39]. Consumers are uncertain about the usefulness of new technology due to the status quo regarding existing technology; hence, they tend to exhibit resistance to new technology [12]. Consumers are used to the existing food delivery systems and naturally are skeptical toward using drone-based food delivery systems given their novelty. Moreover, the drone system is complex, and consumers may perceive that it will not be easy for them to avail drone-based food delivery systems [13]. In previous literature, studies found usage barriers to be a crucial indicator of consumers' resistance intention in different contexts [13], [29], [39]. Based on the above arguments, we hypothesize that

**H1:** *Usage barriers are positively associated with consumers' resistance intention toward drone-based food delivery services.*

### B. Value Barriers

Consumers' perception of value plays a crucial role in framing behavioral intention toward new technology [40]. Perceived value refers to the tradeoff between benefits and costs associated with new technology as perceived by consumers [41]. If consumers perceive that the cost of using new technology is less than its benefits, they tend to resist it [42], [43]. The drone-based food delivery system delivers food quickly to consumers and so offers the benefit of fresh food at a safe temperature [8]. However, consumers have to pay extra money to avail drone-based services [9]. Further, consumers perceive that they need to spend extra effort to learn how to use drone-based services. Hence, they tend to maintain resistance intention toward it. Based on the above arguments, we hypothesize that

**H2:** *Value barriers are positively associated with consumers' resistance intention toward drone-based food delivery services.*

### C. Risk Barriers

Uncertainty related to the performance of new technology is reflected in the risk barrier [13]. Consumers' behavioral intention toward new technology is based on the risk and uncertainty associated with new technology [40]. Financial risk, social risk, performance risk, privacy and security risk, time risk, and psychological risk represent the major elements of consumers' perceived risk [9], [13], [26], [27], [39], [40]. If consumers think that using new technology is risky, they tend to demonstrate negative intentions toward it [44]. In the drone-based food delivery context, consumers are uncertain about the delivery experience [9]. Consumers are unsure how accurately drone food delivery will deliver the food, i.e., whether it will arrive on time and at the correct location [13]. The absence of human touch in drone-based delivery also enhances consumers' psychological risk. Finally, consumers are skeptical about sharing personal and financial information due to the privacy and security risk involved in drone-based food delivery [4]. Based on the above arguments, we hypothesize that

**H3:** *Risk barriers are positively associated with consumers' resistance intention toward drone-based food delivery services.*

### D. Tradition Barriers

A strong connection with existing technology, which refers to the tradition barrier, is one of the major barriers to the usage and adoption of new technology [27]. Over time, consumers develop a sense of familiarity and trust toward existing technology to the extent that they do not want to experiment with new technology [45], [46]. Further, consumers' attachment to existing technology strengthens the status quo, resulting in resistance to new technology [12]. In the food delivery context, if consumers avail food delivery through a delivery person, they become accustomed to it and are henceforth skeptical toward drone-based delivery services [11]. Consumers also think the development of drone food delivery systems will have an adverse impact on the gig economy and employability; hence, they tend to stick with the existing food delivery system [29]. Based on the above arguments, we hypothesize that

**H4:** *Tradition barriers are positively associated with consumers' resistance intention toward drone-based food delivery services.*

### E. Image Barriers

The public image of new technology plays a crucial role in framing consumers' behavioral intention toward technology [29]. If consumers perceive that new technology is not worthy, they tend to resist it [40]. Drone-based food delivery is a new technology, and consumers are not sure about food quality and customer service through this platform, which leads to its negative image in the mind of consumers and their demonstration of resistance. Because drone-based food delivery is new, consumers are uncertain about the performance and exhibit less affective commitment [31]. The absence of human touch also enhances consumers' negative perception of drone-based food

delivery as they are not sure how to connect to the service providers to avail customer service in the case of experiencing service failure [47]. Based on the above arguments, we hypothesize that

**H5:** *Image barriers are positively associated with consumers' resistance intention toward drone-based food delivery services.*

#### F. Experience Barriers

Experience barriers are one of the significant barriers toward new technology adoption [27]. If consumers are not confident about the usage of the technology, they tend to hold negative emotions toward it [48]. Drone-based food delivery is perceived as complex, leading consumers to expect that ordering food using a drone-based food delivery system will not be easy, hence their negative emotion toward it [49]. Quality customer experience enhances customer trust and ultimately enhances positive behavioral intention [31], [50]. In the food delivery context, consumers expect a smooth journey from ordering food to getting it delivered; however, consumers are uncertain about whether drone-based system can offer a satisfying consumer experience [9]. Consumers expect quality food, affordable cost, and a quality customer experience [51], which they are uncertain they will attain, given the novelty of drone-based services, hence their resistance. Based on the above arguments, we hypothesize that

**H6:** *Experience barriers are positively associated with consumers' resistance intention toward drone-based food delivery services.*

#### G. Mediating Effects of Negative Anticipated Emotions

Consumers' anticipated emotions refer to affective responses toward a specific phenomenon [8]. Over time, consumers develop positive or negative perceptions toward environmental cues, significantly affecting behavioral intention [40]. Xie et al. [25] highlighted the role of moral emotions on consumer behavioral intention toward a green product. Jebarajakirthy et al. [40] demonstrated how emotions play a crucial role in the selection of a channel for making a purchase. Positive emotions, such as trust, joy, happiness, interest, and love positively frame consumers' behavioral intention, whereas negative emotions, such as anger, disgust, sadness, and fear enhance consumers' resistance intention [52], [53]. Although the literature sheds light on the role of emotions in consumers' behavioral intention in different contexts, anticipated emotions emerge as a new phenomenon [8]. Anticipated emotions refer to consumers' postbehavioral perceived emotions toward a specific phenomenon [54]. The drone has recently been introduced in the food delivery contexts; hence, it is crucial to know the role of consumers' anticipated emotions. Individuals tend to develop different emotions regarding the future consequences of current choices [8]; these anticipated emotions play a crucial role in framing consumers' responses [55]. In recent literature, attempts were made to investigate the role of anticipated emotions on consumers' behavioral intention toward new technology in different contexts [8], [52], [56], [57], [58], [59]. In the drone food delivery

context, various perceived barriers frame consumers' negative anticipated emotions, leading to consumers' resistance behavior. Uncertainty and risk associated with drone-based food delivery systems enhance consumers' negative anticipated emotions [8]. Further literature has suggested that anticipated negative emotions significantly lead to unfavorable consumer responses [8], [59], [60]. Individuals tend to select the options in which they perceive fewer negative emotions [52]. Similarly, in the drone food delivery context, if consumers develop anticipated negative emotions due to perceived barriers, they tend to exhibit resistance intention. We therefore propose that

**H7:** *Negative anticipated emotions significantly mediate the association among (a) usage barriers, (b) value barriers, (c) risk barriers, (d) tradition barriers, (e) image barriers, (f) experience barriers and consumers' resistance intention toward drone-based food delivery services.*

#### H. Moderating Effect of Customer Innovativeness

Consumer innovativeness means individual tendency to try new things [4], [61]. In the technology adoption literature, Rogers [62] conceptualized innovativeness in the diffusion of innovation (DOI) theory, where he explained innovativeness as an individual tendency to represent favorable intention toward new technology. Literature has suggested that consumers' innovativeness plays a crucial role in framing consumers' behavioral intention toward new technology [4], [63], [64], [65], [66]. Innovative consumers tend to represent less resistance intention toward new technology adoption compared to consumers with low innovativeness [8], [26], [67]. In the drone-based food delivery context, innovative consumers represent a favorable attitude toward drone food delivery as they love to try new things, hence representing less negative anticipation emotions and less resistance intention.

In contrast, low innovative consumers are skeptical about trying new technology, showing resistance toward drone food delivery [8]. Further, while innovative consumers perceive less functional (performance, value, risk, legal) and psychological (tradition, image, experience) barriers hence representing less negative anticipated emotions toward drone food delivery system and tend to show less resistance toward it [8], low innovative consumers perceive several functional and psychological barriers in availing drone food delivery hence representing negative emotions and ultimately representing higher resistance toward drone food delivery [68].

**H8:** *The mediating effects of negative anticipated emotions among (a) usage barriers, (b) value barriers, (c) risk barriers, (d) tradition barriers, (e) image barriers, (f) experience barriers, and toward drone food delivery service are higher for consumers having a low level of innovativeness compared to consumers having a high level of innovativeness.*

TABLE II  
MEASUREMENT MODEL

| Variables and items                                                                                       | FL   |
|-----------------------------------------------------------------------------------------------------------|------|
| <b>Usage Barrier</b>                                                                                      |      |
| I'm worried whether drone food delivery will really perform as well, as it is supposed to                 | 0.67 |
| The thought of using drone food delivery causes me to be concerned about how reliable its service will be | 0.81 |
| I am concerned that drone food delivery will not provide the level of benefits I would be expecting       | 0.85 |
| I believe the drone food delivery are not technically mature so far                                       | 0.79 |
| <b>Value Barrier</b>                                                                                      |      |
| Drone food delivery offers many advantages compared to other channels                                     | 0.78 |
| The use of drone food delivery increases my ability to control food delivery operation                    | 0.75 |
| Drone food delivery offers many advantages compared to other channels                                     | 0.80 |
| The cost of adopting drone food delivery will be more than the benefits receive                           | 0.56 |
| <b>Risk Barrier</b>                                                                                       |      |
| I fear that personal information is not safe over drone food delivery                                     | 0.70 |
| I fear that while I am using drone food delivery, I may pay money to the wrong vendor                     | 0.83 |
| <b>Tradition Barrier</b>                                                                                  |      |
| I find it difficult to contact customer service over drone food delivery                                  | 0.85 |
| I find it difficult to get some information about drone food delivery use                                 | 0.84 |
| I find it difficult to get my problem resolved from drone food delivery service provider                  | 0.70 |
| <b>Image Barrier</b>                                                                                      |      |
| In my opinion, drone food delivery is often too complicated to be useful                                  | 0.90 |
| I have such an image that drone food delivery is difficult to use                                         | 0.92 |
| <b>Experience Barrier</b>                                                                                 |      |
| Drone food delivery often refuses to take responsibility for wrong or delayed deliveries                  | 0.67 |
| Drone food delivery often refuses to take responsibility for the poor quality of food                     | 0.64 |
| It is difficult to make the drone food delivery understand my problem                                     | 0.75 |
| Drone food delivery has very little coordination with the restaurants                                     | 0.78 |
| Customer feedback is not taken seriously by drone food delivery                                           | 0.81 |
| Drone food delivery does not always take responsibility for common service failures                       | 0.86 |
| <b>Personal Innovativeness</b>                                                                            |      |
| Among my peers, I am usually among the first to try new technology                                        | 0.83 |
| If I heard about new technology, I look for ways to experiment with it                                    | 0.61 |
| In general, I am not hesitant to try out new technology                                                   | 0.56 |
| <b>Negative Anticipated Emotion</b>                                                                       |      |
| If I use a drone food delivery services, I will feel ...                                                  |      |
| Disappointed                                                                                              | 0.75 |
| Depressed                                                                                                 | 0.62 |
| Uncomfortable                                                                                             | 0.77 |
| <b>Resistance intention</b>                                                                               |      |
| I do not want to change my current food delivery channel because of drone food delivery adoption          | 0.90 |
| I do not want to change operational decisions because of drone food delivery adoption                     | 0.87 |
| Overall, I do not want to change the current working methods because of drone food delivery adoption      | 0.71 |

Notes: FL= Factor loading

## IV. RESEARCH METHODS

## A. Sample and Data Collection

An online survey was created and administered by an Indian market research company to collect the responses from consumers using online food delivery. To ensure suitable respondents, we used a screening question in the survey questionnaire: "Do you have access to online food delivery services? Only the respondents who positively responded to the above question were allowed to participate in the survey. A total of 1000 consumers were approached, and 427 responses were received. After excluding incompetent responses, a total of 355 (45% female) responses were used for future analysis. We used some checked questions to check the sincerity of the respondents, and we discarded the responses that do not qualify the check question. Most respondents belong to the 21–30 years (29%) and the 31–40 (26 percent) age group. A total of 22% of respondents belong to the 41–50 age group, and the rest of the respondents belong to the above 50 age brackets. Educationwise, the majority of the respondents have a graduate degree (38%), followed by 28% who hold a postgraduate degree, with a minority having senior secondary school as the highest qualification degree (16%). Incomewise, 16% of respondents had a monthly income below USD 200, 22% had between USD 201 – USD 400, 19% earned between USD 400–USD 600, and the remaining earned above USD 600 (see Table II).

## B. Scale Development

We adapted the validated scales from the literature to develop measurement items for the study constructs. Specifically, the

TABLE III  
DISCRIMINANT VALIDITY ANALYSIS

| Variables                          | AVE  | CR   | $\alpha$ | UB          | VB          | RB          | TB          | IB          | EB          | PI          | NAE         | RI          |
|------------------------------------|------|------|----------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Usage Barrier (PB)                 | 0.61 | 0.86 | 0.86     | <b>0.78</b> |             |             |             |             |             |             |             |             |
| Value Barrier (VB)                 | 0.51 | 0.79 | 0.77     | 0.66        | <b>0.71</b> |             |             |             |             |             |             |             |
| Risk Barrier (RB)                  | 0.59 | 0.74 | 0.74     | 0.45        | 0.32        | <b>0.77</b> |             |             |             |             |             |             |
| Tradition Barrier (TB)             | 0.63 | 0.84 | 0.83     | 0.67        | 0.71        | 0.34        | <b>0.79</b> |             |             |             |             |             |
| Image Barrier (IB)                 | 0.82 | 0.91 | 0.91     | 0.62        | 0.54        | 0.44        | 0.55        | <b>0.91</b> |             |             |             |             |
| Experience Barrier (EB)            | 0.57 | 0.88 | 0.88     | 0.59        | 0.61        | 0.31        | 0.67        | 0.57        | <b>0.75</b> |             |             |             |
| Personal Innovativeness (PI)       | 0.51 | 0.71 | 0.71     | 0.64        | 0.69        | 0.29        | 0.64        | 0.46        | 0.58        | <b>0.71</b> |             |             |
| Negative Anticipated Emotion (NAE) | 0.51 | 0.75 | 0.74     | 0.63        | 0.64        | 0.38        | 0.55        | 0.70        | 0.66        | 0.41        | <b>0.71</b> |             |
| Resistance intention (RI)          | 0.69 | 0.87 | 0.85     | 0.73        | 0.65        | 0.39        | 0.73        | 0.66        | 0.73        | 0.57        | 0.45        | <b>0.83</b> |

Notes: AVE= Average variance extracted, CR= Composite reliability,  $\alpha$ = Cronbach's alpha coefficient, Diagonal value indicates the square root of AVE of individual latent construct

study constructs and the sources of the items used to measure these constructs are usage barriers (four items), value barriers (four items), risk barriers (two items), image barriers (two items), tradition barriers (three items), and experience barriers (six items) from Kaur et al. [13]; anticipated negative emotions (three items) from Hwang et al. [8]; innovativeness (three items) from Shankar et al. [4] and resistance intention (three items) from Shankar and Nigam [14]. We used a five-point Likert scale (1-strongly disagree to 5-strongly agree) to measure study constructs.

## V. ANALYSIS AND RESULTS

## A. Reliability and Validity of the Measurement Model

We conducted a confirmatory factor analysis (CFA) using AMOS 27 software to assess the reliability and validity of the measurement items used in the survey questionnaire [69]. The findings of CFA presented in Table III show that Cronbach's alpha for all variables is above 0.7 to ensure consistency [69]. We examined the convergent and discriminant validity of the measurement scale [69]. The values of average variance extracted (AVE) and composite reliability (CR) for all the constructs are above 0.5 and 0.7, respectively, thereby confirming the convergent validity of the scale [67]. The findings of discriminant validity show that the square root of AVE for each construct is greater than the correlation coefficients of the corresponding constructs, confirming the discriminant validity of the measurement constructs [69]. The variance inflation factors (VIF) for the study variables were calculated to test for multicollinearity. The VIF values ranged from 1.19 to 2.79, which were less than the threshold value of 3, suggesting that all the variables considered in the study were free from any issues related to multicollinearity [66].

## B. Common Method Bias (CMB)

We performed several statistical and nonstatistical techniques to assess the presence of CMB. As a nonstatistical measure, clarity of the survey items and the confidentiality of the respondents were ensured to confirm the absence of common method variance [71]. As a statistical measure, to assess the probability of CMB, Harman's one-factor test was performed. According to the analysis, one factor items explained 21.54% of the variance, which is less than the suggested threshold value of 50%. This demonstrates that the study was free from the risk of CMB [71].

## C. Hypotheses Testing

Results highlighted in Table IV indicate that usage barrier

TABLE IV  
PATH ANALYSIS

| Path                                                  | $\beta$  | SE    |
|-------------------------------------------------------|----------|-------|
| Usage Barrier $\rightarrow$ Resistance Intention      | 0.307*** | 0.110 |
| Value Barrier $\rightarrow$ Resistance Intention      | -0.039ns | 0.086 |
| Risk Barrier $\rightarrow$ Resistance Intention       | 0.017ns  | 0.052 |
| Tradition Barrier $\rightarrow$ Resistance Intention  | 0.24***  | 0.077 |
| Image Barrier $\rightarrow$ Resistance Intention      | 0.176*** | 0.047 |
| Experience Barrier $\rightarrow$ Resistance Intention | 0.304**  | 0.096 |
| Age $\rightarrow$ Resistance Intention                | 0.007ns  | 0.051 |
| Gender $\rightarrow$ Resistance Intention             | -0.023ns | 0.033 |

Notes: \*\*\*  $p < 0.001$ ; \*\*  $p < 0.01$ ; ns = not significant

TABLE V  
MEDIATION ANALYSIS

| Hypothesis                                            | Effects | BootSE | Bootstrap 95% CIs | Mediation |
|-------------------------------------------------------|---------|--------|-------------------|-----------|
|                                                       |         |        | Lower Upper       |           |
| Usage Barrier $\rightarrow$ NAE $\rightarrow$ RI      | .0228   | .0145  | .001 .0572        | Yes       |
| Value Barrier $\rightarrow$ NAE $\rightarrow$ RI      | .0077   | .0117  | -.0144 .0331      | No        |
| Risk Barrier $\rightarrow$ NAE $\rightarrow$ RI       | -.0005  | .0057  | -.0126 .011       | No        |
| Tradition Barrier $\rightarrow$ NAE $\rightarrow$ RI  | .0692   | .0324  | .0085 .1353       | Yes       |
| Image Barrier $\rightarrow$ NAE $\rightarrow$ RI      | .0253   | .0147  | .0013 .0588       | Yes       |
| Experience Barrier $\rightarrow$ NAE $\rightarrow$ RI | .0179   | .0107  | .0015 .0428       | Yes       |

Notes: NAE = Negative Anticipated Emotion, RI = Resistance Intention

( $\beta = 0.31, p < 0.001$ ), tradition barrier ( $\beta = 0.24, p < 0.001$ ), image barrier ( $\beta = 0.18, p < 0.001$ ) and experience barrier ( $\beta = 0.30, p < 0.01$ ) have significant positive impacts on resistance intention toward drone-based food delivery. Hence, H1, H4, H5, and H6 are supported. However, the results show that value barrier ( $\beta = -0.04, p > 0.05$ ) and risk barrier ( $\beta = 0.02, p > 0.05$ ) did not have a significant effect on inertia, thereby indicating that H2 and H3 are not supported. The structural model explained 71.7% of variance for resistance intention toward drone-based food delivery services. None of the control variables (age and gender) significantly affect resistance intention toward drone-based food delivery services.

We used Model 4 in Process Macro, Hayes [69] with a non-parametric bootstrapping method for 5000 resamples to examine mediation effects of negative anticipated emotions between barriers and resistance intention toward drone-based food delivery. Results highlighted in Table V show that negative anticipated emotions significantly mediate the association among usage barriers (Indirect effect = .022, LLCI = .001, ULCI = .0572), tradition barriers (Indirect effect = .069, LLCI = .0085, ULCI = .1353), image barriers (Indirect effect = .025, LLCI = .0013, ULCI = .0588), experience barriers (Indirect effect = .018, LLCI = .0015, ULCI = .0428), and resistance intention toward drone-based food delivery services. Hence, H7a, H7d, H7e, and H7f are supported. In comparison, negative anticipated emotions did not mediate the association among value barriers (Indirect effect = .007, LLCI = -.0144, ULCI = .0331), risk barriers (Indirect effect = .025, LLCI = -.0126, ULCI = .011), and resistance intention toward drone-based food delivery services. Hence, H7b and H7c are not supported.

To test the moderated mediation hypotheses, we used bootstrap estimates from 5000 samples using Model 7 in Process Macro [72]. Table VI shows that consumers' innovativeness negatively moderated the mediating effect of negative anticipated emotions in the association among usage barrier (Effect = -0.381;

TABLE VI  
MODERATED-MEDIATION ANALYSIS

| Paths                                                 | Effects | Bootstrap 95% CIs |        |        |
|-------------------------------------------------------|---------|-------------------|--------|--------|
|                                                       |         | BootSE            | Lower  | Upper  |
| Usage Barrier $\rightarrow$ NAE $\rightarrow$ RI      | -.0381  | .0178             | -.0735 | -.003  |
| Tradition Barrier $\rightarrow$ NAE $\rightarrow$ RI  | .0098   | .0266             | -.0348 | .0696  |
| Image Barrier $\rightarrow$ NAE $\rightarrow$ RI      | -.0316  | .0148             | -.0602 | -.0011 |
| Experience Barrier $\rightarrow$ NAE $\rightarrow$ RI | -.0606  | .0212             | -.1036 | -.0185 |

Notes: NAE = Negative Anticipated Emotion, RI = Resistance Intention

LLCI = -.0735, ULCI = -.003), image barrier (Effect = -0.316; LLCI = -.0602, ULCI = -.0011), experience barrier (Effect = -0.606; LLCI = -.1036, ULCI = -.0185), and resistance intention toward drone-based food delivery services, thereby supporting H8a, H8e, and H8f. However, mediating effect of negative anticipated emotions on the association between tradition barrier (Effect = .0098; LLCI = -.0348, ULCI = .0696) and resistance intention toward drone-based food delivery services do not vary at different levels of consumer innovativeness. Hence, H8d is not supported.

## VI. DISCUSSION

In recent years, in the hospitality industry especially, the food delivery services industry has used advanced technology to deliver food to consumers. Drone food delivery is one of the innovative technologies which is attracting significant attention for delivering food. Although, drone-based food delivery helps in delivering food quickly to remote area, consumers' adoption intention toward drone-based food delivery is not encouraging. Hence, it is crucial to examine consumers' resistance intention toward drone-based food delivery systems. This study proposes a framework using innovation resistance theory to examine consumers' resistance intention toward drone-based food delivery. We have examined the role of functional and psychosocial barriers in framing consumers' resistance intention. The study also examines how anticipated negative emotions mediate the association between barriers and resistance intention and how mediation effects vary at different levels of consumers' innovativeness. Consistent with previous findings, the results indicate that usage barriers, tradition barriers, image barriers, and experience barriers are positively associated with resistance intention [13], [74]. Consistent with the findings of the previous studies in other contexts [17], the results indicated that usage barriers have a significant positive effect on resistance behavior. As drone-based food delivery system is new, consumers are not familiar with drones' characteristics and use for food delivery. Consumers are also not sure how using drone-based delivery would be beneficial. Further, consumers perceive that ordering food using a drone-based food delivery system might be complex and hence tend to demonstrate resistance intention toward it.

Consistent with the findings of previous research [74] in other contexts, results indicated that traditional barriers significantly enhance consumers' resistance intention toward drone-based food delivery systems. With respect to these systems, consumers are not sure how they will connect to the customer service team if they receive food of unsatisfactory quality. They are also not

confident in the ability of drone-based food delivery service providers to resolve the issues in case of service failure; hence they tend to resist it. Further, due to less awareness, consumers perceive drone-based food delivery systems as complicated, and hence image barriers play a crucial role in framing consumers' resistance intention toward drone-based food delivery services. The above finding is consistent with the findings of a previous study [45]. In line with the findings of Kaur et al. [13], the results also indicated that experience barriers are significantly associated with resistance intention. Consumers are not confident about what the service experience would be like.

Further, consumers are also confused about who will take responsibility in case of service failure; hence they tend to resist drone-based food delivery. Contradictory to the findings of previous studies by Laukkanen [45] and Sivathanu [73], results indicated that value barriers have no significant effect on resistance intention. Although consumers are not sure about the performance of drone-based food delivery, the cost of availing services is convenient compared to other means of food delivery. Hence, consumers find drone-based food delivery valuable. Results also indicated that risk barriers have no significant effect on resistance intention as drone-based food delivery services are provided by existing food delivery service providers only; hence, consumers are sure of privacy and security of the personal and financial information shared by consumers. Consequently, consumers do not feel insecure about using drone-based food delivery services.

Furthermore, results indicated that consumers' demographic (age and gender) has no significant effect on consumers' resistance intention toward drone-based food delivery services. Hence, a standardized delivery system is sufficient to enhance the reach of consumers' experience. Marketers should therefore focus on mass market strategies in the drone-based food delivery context.

Consistent with Hwang et al. [8], results suggested that negative anticipated emotions play a crucial role in framing consumers' resistance intention toward drone-based food delivery services. If consumers perceive barriers in adopting drone food services, they tend to show negative emotions, such as disappointment, depression, and discomfort and ultimately represent resistance to drone-based food delivery systems. Finally, results indicated that for innovative consumers, different functional and psychological barriers have less impact on consumers' perceived negative anticipated emotions and consumers' resistance intention toward drone-based food delivery services compared to less innovative consumers. Innovative consumers show less negative anticipated emotions toward new technology and hence display less resistance intention toward new technology. In comparison, low innovative consumers show inertia toward the existing system and resist using drone food delivery services [4].

## VII. IMPLICATIONS

### A. Theoretical Implications

This study has several academic implications. First, in literature studies focused on online food delivery [75], [76], [77] and drone-based delivery services recently received the attention of

the researchers, and the majority of previous studies attempted to investigate supply chain-related issues [78], [79], [80]. However, little effort was made to examine consumers' acceptance of drone-based delivery services including scant efforts to investigate consumers' resistance intention in food delivery contexts. This study uniquely contributes to the food delivery-related literature by investigating the effects of functional and psychological barriers on consumers' resistance intention toward drone-based food delivery services. We developed a comprehensive model with the underpinnings of innovation resistance theory and empirically validated it in the context of drone-based food delivery services. The model proposed six types of barriers (i.e., usage, value, risk, tradition, image, and experience) and negative anticipated emotions as mediators. The study also examined how mediating effects of negative anticipated emotions vary at different levels of consumers' innovativeness.

Second, this study extended the consumers' emotions-related literature by investigating the mediating effects of negative anticipated emotions on the association among barriers and resistance intention. Third, the study also contributes to consumers' innovativeness literature as our study shows some boundary conditions for mediation effects of negative anticipated emotions on the association among barriers and resistance intention. We conducted a moderated mediation analysis using personal innovativeness as moderator.

Finally, in existing literature, IRT was used to investigate individual resistance intention in different contexts [13], [74]. This study enriches IRT literature by using it as an underpinning theory to investigate consumers' resistance intention toward drone-based food delivery services.

This study enriches the IRT literature in several ways. First, we have extended IRT by adding experience barriers along with the several types of barriers used in existing literature. Second, we have applied consumers' resistance toward innovation in emerging markets, such as India. In India, marketers have started using or are planning to use drones to deliver products, however, they are uncertain about consumers' response. Hence, it is crucial to know how consumers will react. Furthermore, in existing literature, the majority of the studies investigated how different barriers and risks affect consumers' resistance behavior. However, consumers' resistance behavior is a complex process which is affected by several intervening variables. This study contributes to the IRT literature by investigating how the effects of different barriers are mediated by negative anticipated emotions and moderated by consumers' innovativeness. Finally, this study also confirms the suitability of the IRT in investigating consumers' resistance behaviors, given that this study's framework based on IRT explained 71.7% of variance for resistance intention toward drone-based food delivery services.

### B. Managerial Implications

Based on our study's findings, we here provide several insightful recommendations to food delivery service providers. Since our findings show that usage barriers, tradition barriers, image barriers, and experience barriers drive consumers' resistance intention toward drone-based food delivery services, we

recommend that marketers consider these four barriers when designing such services. More specifically, to reduce usage barriers, marketers should provide a solution that is easy to use and understand. The organization should also provide discounts to motivate consumers to use drone-based food delivery services. Marketers should also ensure zero defect services to reduce consumers' sense of risk. Marketers should also ensure the quality and quantity of the food on time.

Further, marketers should ensure smooth and efficient after services to reduce tradition barriers. The organizations should ensure effective solutions at all hours of operation in case of a failure situation. Further, there must be several means to connect with marketers if consumers face issues. Marketers should also attempt to reduce consumers' image barriers by making consumers aware of drone-based food delivery services. Marketers should also use drones to deliver food in remote areas without failure. Finally, to reduce consumers' experience, marketers should provide a better moment of truth at each touchpoint while delivering food using drones.

The study also suggested that negative anticipated emotions play a crucial role in framing consumers' resistance intention toward drone-based food delivery services. Hence, the organization should provide utilitarian and hedonic values to reduce negative emotions. Finally, results suggested that consumers' innovativeness has negative moderating effects; hence, marketers should consider consumers' innovativeness as a criterion for segmentation.

## VIII. CONCLUSION

Although this study has multiple academic and practical implications, it has a few limitations that provide avenues for future research. First, our proposed framework was validated in the food delivery context. However, future research can be conducted to make the model more generalizable to the broader service sector. Second, we investigated our model based on cross-sectional data. Since human behavior changes over time, the model could provide more exciting findings if data from multiple time periods are used in testing the proposed framework. Third, the current study examined the proposed model using the data collected from one country, India, and so to generalize the findings better, the proposed model can be examined using the data collected from multiple countries [81]. Moving forth, food delivery service providers are taking first steps in adopting drone technology, hence it is interesting to investigate how consumers will respond toward AI and machine learning-enabled drone-based food delivery systems in future. Furthermore, consumers' resistance toward drone-based food delivery could be due to less awareness about technology in the case of Indian rural consumers. In future, studies might examine the role of technological awareness. Fifth, we have used negative anticipated emotions as a mediator in the proposed framework. In future, studies might examine the effect of relevant mediators, such as distrust, inertia, attitude, and perceived values in the existing framework. Moreover, studies might advance the proposed framework by investigating the moderating effects of involvement, perceived risk, social norms, and environmental consciousness for deeper understanding of the phenomenon.

Finally, the scope of this study is limited to investigating consumers' resistance intention; future studies might examine both adoption and resistance behavior using the dual-factor model to enrich the existing literature.

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